

# What this PDF contains

This document focuses on the current microservice layout, the most important runtime flows, and the logical entity relationships used by the backend. Cross-service references are shown as logical IDs because the services own separate databases.

## Included pages

- 1. Deployment UML view: gateway, discovery, services, databases, Kafka, Redis, and providers.
- 2. Interaction UML view: real-time chat, translation-credit flow, and payment top-up flow.
- 3. ER view: key JPA entities and their logical relationships across services.

## How to read the diagrams

Blue arrows show synchronous HTTP or Feign calls.  
Amber arrows show Kafka or event-driven communication.  
Gray dashed arrows show service discovery or logical references.  
Green database boxes mean persistent state. Teal boxes show service components.

## Source evidence used

README.md and docs/PRODUCTION\_DOCUMENTATION.md for ports, runtime topology, and flows.  
application.yaml files for datasource, Redis, Kafka, and Eureka wiring.  
Entity classes in auth-service, room-service, message-service, notification-service, and payment-service.

# Service inventory

### gateway-service

Port 9001  
Entry point for HTTP and WebSocket

### eureka-server

Port 9000  
Service registry and discovery

### auth-service

Port 9002  
Users, contacts, JWT, credits

### room-service

Port 9003  
Rooms and room members

### message-service

Port 9004  
Message persistence and search

### notification-service

Port 9007  
Unread notifications

### presence-service

Port 9012  
Redis-backed presence

### translation-service

Port 9013  
Gemini and LibreTranslate wrapper

### websocket-service

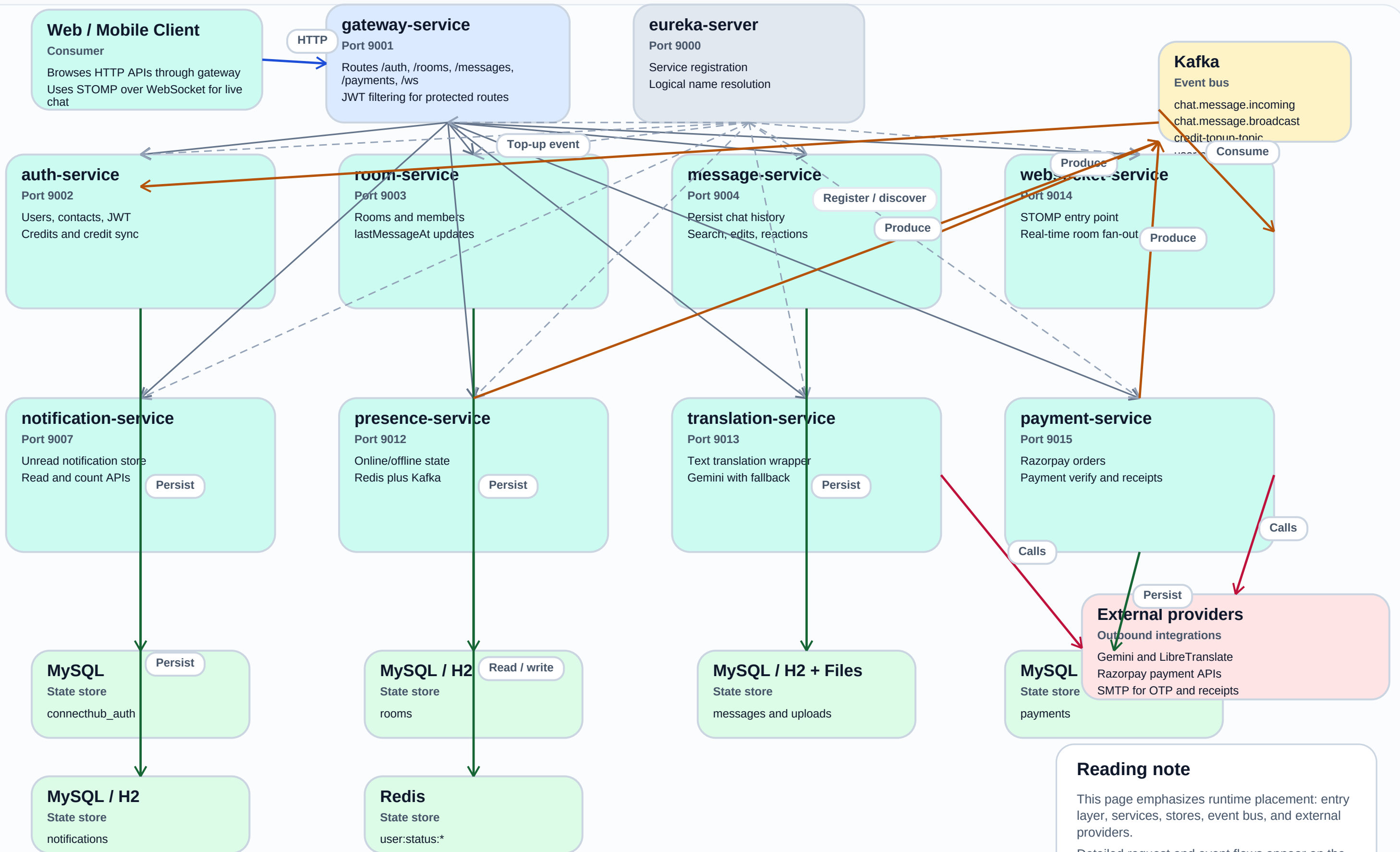
Port 9014  
STOMP and real-time fan-out

### payment-service

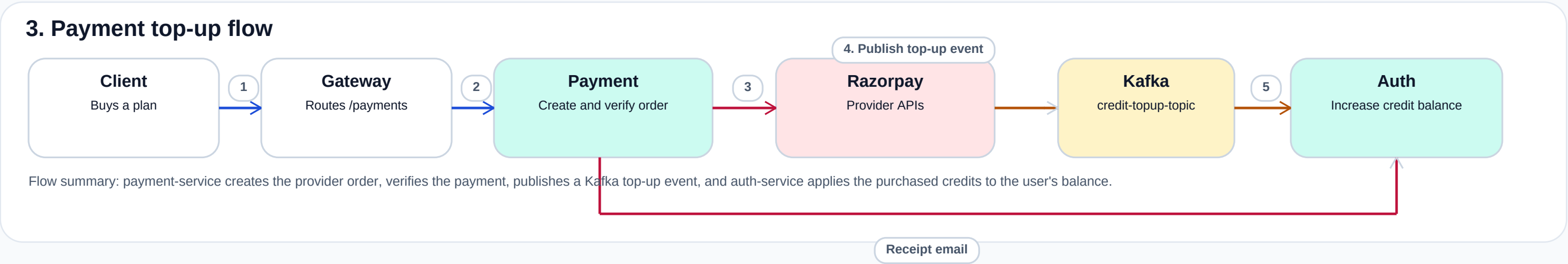
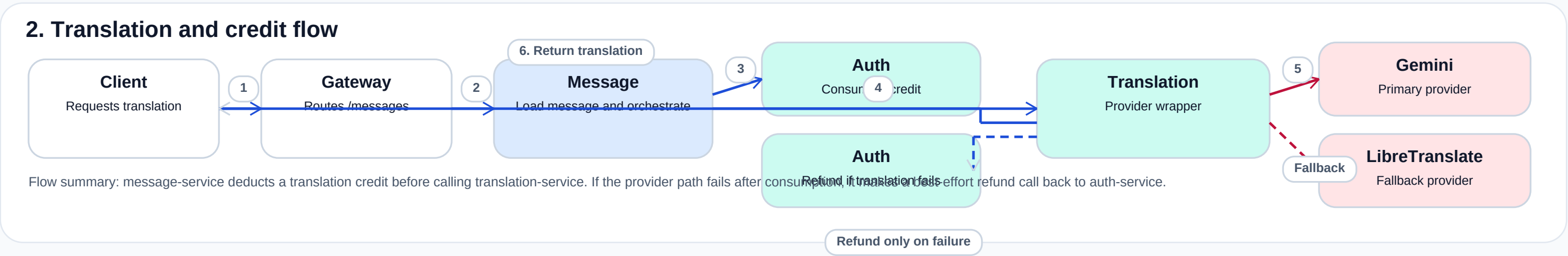
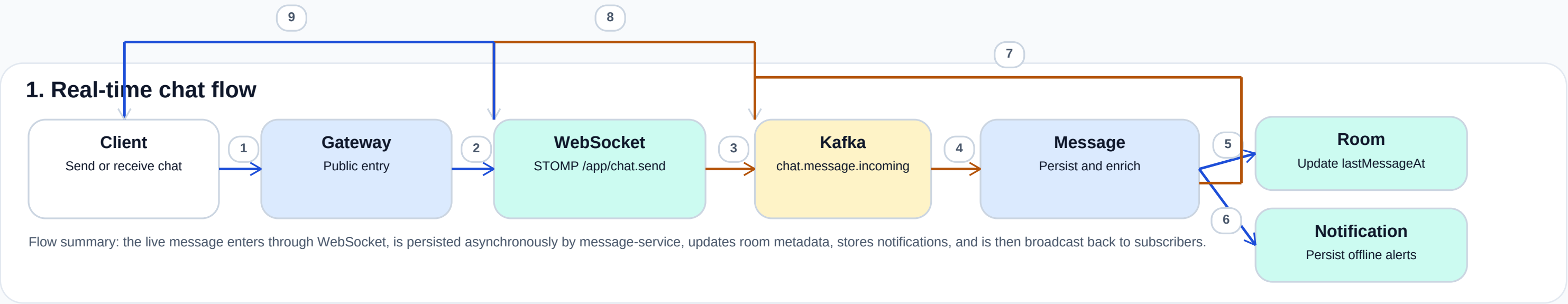
Port 9015  
Razorpay orders and credit sync

# UML Deployment View

High-level component diagram of the ConnectHub backend runtime topology.



Three important business flows are separated so each one is easy to follow.



Logical entity relationships across the ConnectHub services. Only key fields are shown to keep the page readable.

